

MLE, EM Algorithm, and Bootstrap Inference

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Maximum Likelihood Estimation

MLE

MAXIMUM LIKELIHOOD ESTIMATION

“Statistical” setup:

$$z \sim f(z \mid \theta)$$

MLE estimator of θ is

$$\hat{\theta} = \arg \max_{q \in \Theta} \mathcal{L}(z_1, \dots, z_n, q)$$

where $\mathcal{L}(z_1, \dots, z_n, \theta)$ is the likelihood function, which is the joint density/mass of observations $z_1, \dots, z_n$ in the sample.

Suppose $z_1, \dots, z_n$ is a random sample, then

$$\hat{\theta} = \arg \max_{q \in \Theta} \prod_{i=1}^n f(z_i \mid q) = \arg \max_{q \in \Theta} \frac{1}{n} \sum_{i=1}^n \log f(z_i \mid q)$$

where

$$\sum_{i=1}^n \log f(z_i \mid q) = \ell(z_1, \dots, z_n, q) = \ell_n(q)$$

is the loglikelihood function.

Why does MLE work? **Identification condition. Lemma (Information inequality).**

Suppose that

$$\Pr\{f(z \mid q_1) \neq f(z \mid q_2)\} > 0$$

for any $q_1 \neq q_2$. Then $E[\log f(z \mid q)]$ is uniquely maximized at θ .

Definition. Score function is

$$s(z, q) = \frac{\partial \log f(z | q)}{\partial q}.$$

Definition. True score is

$$s(z, \theta).$$

Lemma (Zero expected score).

$$E[s(z, \theta)] = 0.$$

Proof. Consider the F.O.C. to the optimization problem

$$\theta = \arg \max_{q \in \Theta} E[\log f(z | q)].$$

Goldberger's idea: apply the analogy principle to “zero expected score”:

$$\frac{1}{n} \sum_{i=1}^n s(z_i, \hat{\theta}_{ZES}) = 0.$$

Question: How does $\hat{\theta}_{ZES}$ relate to MLE $\hat{\theta}$?

Asymptotic properties of MLE

Definition. Information matrix is a $k \times k$ matrix

$$\mathcal{I}(q) = -E \left[\frac{\partial s(z, q)}{\partial q'} \right].$$

Lemma (Information matrix equality).

True information matrix satisfies

$$\mathcal{I}(\theta) = E[s(z, \theta)s(z, \theta)'].$$

Global identification condition.

$$\Pr\{f(z \mid q_1) \neq f(z \mid q_2)\} > 0 \quad \text{for any } q_1 \neq q_2.$$

Local identification condition.

True information matrix $\mathcal{I}(\theta)$ is non-singular.

- MLE is consistent:

$$\hat{\theta} \xrightarrow{P} \theta.$$

- MLE is asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V_\theta).$$

- Asymptotic variance is

$$\begin{aligned} V_\theta &= E \left[\frac{\partial s(z, \theta)}{\partial q'} \right]^{-1} E[s(z, \theta)s(z, \theta)'] E \left[\frac{\partial s(z, \theta)}{\partial q'} \right]^{-1} \\ &= (-\mathcal{I}(\theta))^{-1} \mathcal{I}(\theta) (-\mathcal{I}(\theta))^{-1} = \mathcal{I}(\theta)^{-1}. \end{aligned}$$

Note: “sandwich” collapses!

Pivotization

Asymptotic variance is sandwich

$$V_{\theta} = \mathcal{I}(\theta)^{-1},$$

where

$$\mathcal{I}(\theta)^{-1} = -E \left[\frac{\partial s(z, \theta)}{\partial q'} \right]^{-1} \quad (\text{from definition})$$

$$\mathcal{I}(\theta)^{-1} = E[s(z, \theta)s(z, \theta)']^{-1} \quad (\text{from Information matrix equality})$$

and before sandwich collapses,

$$\mathcal{I}(\theta)^{-1} = E \left[\frac{\partial s(z, \theta)}{\partial q'} \right]^{-1} E[s(z, \theta)s(z, \theta)'] E \left[\frac{\partial s(z, \theta)}{\partial q'} \right]^{-1}.$$

Three estimators then:

$$\hat{V}_{\theta}^{(1)} = \left(-\frac{1}{n} \sum_{i=1}^n \frac{\partial s(z_i, \hat{\theta})}{\partial q'} \right)^{-1},$$

$$\hat{V}_{\theta}^{(2)} = \left(\frac{1}{n} \sum_{i=1}^n s(z_i, \hat{\theta})s(z_i, \hat{\theta})' \right)^{-1},$$

$$\hat{V}_{\theta}^{(3)} = \left(\frac{1}{n} \sum_{i=1}^n \frac{\partial s(z_i, \hat{\theta})}{\partial q'} \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n s(z_i, \hat{\theta})s(z_i, \hat{\theta})' \right) \left(\frac{1}{n} \sum_{i=1}^n \frac{\partial s(z_i, \hat{\theta})}{\partial q'} \right)^{-1}.$$

Asymptotic variance estimators

We have three estimators:

$$\hat{V}_\theta^{(1)} = \left(-\frac{1}{n} \sum_{i=1}^n \frac{\partial s(z_i, \hat{\theta})}{\partial q'} \right)^{-1},$$

$$\hat{V}_\theta^{(2)} = \left(\frac{1}{n} \sum_{i=1}^n s(z_i, \hat{\theta}) s(z_i, \hat{\theta})' \right)^{-1},$$

$$\hat{V}_\theta^{(3)} = \left(\frac{1}{n} \sum_{i=1}^n \frac{\partial s(z_i, \hat{\theta})}{\partial q'} \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n s(z_i, \hat{\theta}) s(z_i, \hat{\theta})' \right) \left(\frac{1}{n} \sum_{i=1}^n \frac{\partial s(z_i, \hat{\theta})}{\partial q'} \right)^{-1}.$$

- Consistent? Yes if ULLN holds for all averages.
- Symmetric? Yes.
- Positive definite? Yes.
- Which is better?

Example: Bernoulli distribution

Let

$$z_1, \dots, z_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(\theta), \quad 0 < \theta < 1.$$

Density and logdensity:

$$f(z \mid q) = q^z (1 - q)^{1-z},$$

$$\log f(z \mid q) = z \log q + (1 - z) \log(1 - q).$$

Score function:

$$s(z, q) = \frac{z}{q} - \frac{1 - z}{1 - q} = \frac{z - q}{q(1 - q)}.$$

Loglikelihood:

$$\ell_n(q) = \sum_{i=1}^n [z_i \log q + (1 - z_i) \log(1 - q)].$$

ZES estimator:

$$0 = \sum_{i=1}^n \left(\frac{z_i}{\hat{\theta}_{ZES}} - \frac{1 - z_i}{1 - \hat{\theta}_{ZES}} \right)$$

$$\Rightarrow \hat{\theta}_{ZES} = \frac{1}{n} \sum_{i=1}^n z_i \equiv \bar{z}_n.$$

Example: Bernoulli distribution (continued)

ML estimator:

$$\hat{\theta} = \arg \max_{q \in (0,1)} \sum_{i=1}^n [z_i \log q + (1 - z_i) \log(1 - q)].$$

F.O.C.:

$$\begin{aligned} 0 &= \sum_{i=1}^n \left(\frac{z_i}{\hat{\theta}} - \frac{1 - z_i}{1 - \hat{\theta}} \right) \\ \Rightarrow \hat{\theta} &= \frac{1}{n} \sum_{i=1}^n z_i = \hat{\theta}_{ZES}. \end{aligned}$$

We are lucky that ML estimator exists in closed (explicit) form.

Information matrix:

$$\mathcal{I}(\theta) = E \left[\frac{(z - \theta)^2}{\theta^2(1 - \theta)^2} \right] = \frac{1}{\theta(1 - \theta)}.$$

Asymptotic variance:

$$V_{\theta} = \theta(1 - \theta).$$

Estimate of asymptotic variance:

$$\hat{V}_{\theta} = \hat{\theta}(1 - \hat{\theta}) = \bar{z}_n(1 - \bar{z}_n).$$

Conditional MLE

“Econometric” setup:

$$y \mid x \sim f(y \mid x, \theta)$$

where $f(y \mid x, \theta)$ is density/mass of y conditional on x .

$$f(y, x \mid \theta) = f(y \mid x, \theta)f(x \mid \theta).$$

Assumption (distribution of x is variation-free):

$$f(x \mid \theta)$$

does not depend on θ .

Then

$$\begin{aligned}\hat{\theta}^J &= \arg \max_{q \in \Theta} \frac{1}{n} \sum_{i=1}^n \log f(y_i, x_i \mid q) \\ &= \arg \max_{q \in \Theta} \left\{ \frac{1}{n} \sum_{i=1}^n \log f(y_i \mid x_i, q) + \frac{1}{n} \sum_{i=1}^n \log f(x_i) \right\} = \hat{\theta}^C.\end{aligned}$$

And all theory still applies to $\hat{\theta}^C$.

Example: Normal linear regression: MLE

Consider linear mean regression with normal specification of conditional density:

$$y \mid x \sim N(x'\beta, \sigma^2).$$

with

$$q = (b', s^2)', \quad \theta = (\beta', \sigma^2)'.$$

Conditional logdensity is

$$\log f(y \mid x, q) = \text{const} - \frac{1}{2} \log s^2 - \frac{(y - x'b)^2}{2s^2}.$$

So MLE estimator is

$$\begin{pmatrix} \hat{\beta} \\ \hat{\sigma}^2 \end{pmatrix} = \arg \max_{b, s^2} \sum_{i=1}^n \left\{ -\frac{1}{2} \log s^2 - \frac{(y_i - x_i'b)^2}{2s^2} \right\}.$$

Hence

$$\begin{pmatrix} \hat{\beta} \\ \hat{\sigma}^2 \end{pmatrix} = \begin{pmatrix} \text{OLS estimator} \\ \text{OLS residual variance} \end{pmatrix}$$

Example: Normal linear regression: Information matrix

Conditional logdensity is

$$\log f(y \mid x, q) = \text{const} - \frac{1}{2} \log s^2 - \frac{(y - x' b)^2}{2s^2}.$$

Conditional score function is

$$s(y \mid x, q) = \begin{pmatrix} \frac{y - x' b}{s^2} x \\ -\frac{1}{2s^2} + \frac{(y - x' b)^2}{2s^4} \end{pmatrix}.$$

Also,

$$\frac{\partial s(y \mid x, q)}{\partial q'} = \begin{pmatrix} -\frac{xx'}{s^2} & -\frac{y - x' b}{s^4} x \\ -\frac{y - x' b}{s^4} x' & \frac{1}{2s^4} - \frac{(y - x' b)^2}{s^6} \end{pmatrix}.$$

For information matrix, using definition,

$$\begin{aligned} \mathcal{I}(\theta) &= -E \left[\frac{\partial s(y \mid x, \theta)}{\partial q'} \right] \\ &= \frac{1}{\sigma^2} \begin{pmatrix} E[xx'] & 0 \\ 0' & \frac{1}{2\sigma^2} \end{pmatrix}. \end{aligned}$$

Example: Normal linear regression: MLE asymptotics

Information matrix:

$$\mathcal{I}(\theta) = \frac{1}{\sigma^2} \begin{pmatrix} E[\mathbf{x}\mathbf{x}'] & 0 \\ 0' & \frac{1}{2\sigma^2} \end{pmatrix}.$$

Asymptotic properties:

$$\begin{pmatrix} \hat{\beta} \\ \hat{\sigma}^2 \end{pmatrix} \xrightarrow{p} \begin{pmatrix} \beta \\ \sigma^2 \end{pmatrix}$$

and

$$\sqrt{n} \begin{pmatrix} \hat{\beta} - \beta \\ \hat{\sigma}^2 - \sigma^2 \end{pmatrix} \xrightarrow{d} N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2 \mathbf{Q}_{\mathbf{x}\mathbf{x}}^{-1} & 0 \\ 0' & 2\sigma^4 \end{pmatrix} \right].$$

Notable result.

ML estimators of $\hat{\beta}$ and $\hat{\sigma}^2$ are asymptotically independent.

ML asymptotic tests

“Trinity” (“triad”) of ML asymptotic tests.

Consider simple null hypothesis

$$H_0 : \theta = \theta_0.$$

1. Wald test.

Recall that

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \mathcal{I}(\theta)^{-1}).$$

Then, pivotizing and taking “square”,

$$W = n(\hat{\theta} - \theta_0)' \hat{\mathcal{I}}(\hat{\theta})(\hat{\theta} - \theta_0) \xrightarrow{d}_{H_0} \chi_k^2$$

2. Likelihood ratio.

Taylor-expand loglikelihood:

$$\ell_n(q) = \sum_{i=1}^n \log f(z_i \mid q).$$

To second order around $\hat{\theta}$

$$LR = 2 \left(\ell_n(\hat{\theta}) - \ell_n(\theta_0) \right) \xrightarrow{d}_{H_0} \chi_k^2$$

ML asymptotic tests

3. Score (Lagrange multiplier) test.

Recall the ZES rule:

$$E[s(z, \theta_0)] = 0.$$

Under H_0 , by CLT and information matrix equality,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n s(z_i, \theta_0) \xrightarrow{d_{H_0}} N(0, \mathcal{I}(\theta_0)).$$

Pivoting and taking “square”,

$$LM = \frac{1}{n} \left(\sum_{i=1}^n s(z_i, \theta_0) \right)' \hat{\mathcal{I}}(\theta_0)^{-1} \left(\sum_{i=1}^n s(z_i, \theta_0) \right) \xrightarrow{d_{H_0}} \chi_k^2$$

ML under density misspecification

What if the true conditional density $g(y | x)$ is different from specified family $f(y | x, \theta)$?

$$\hat{\theta} = \arg \max_{q \in \Theta} \frac{1}{n} \sum_{i=1}^n \log f(y_i | x_i, q).$$

Is $\hat{\theta}$ still consistent for θ ?

It may or it may not.

Result. QML (in)consistency

- Some densities do provide QML consistency even when true density is misspecified: normal, Poisson, exponential and gamma, Bernoulli, etc.
- Some densities fail QML consistency when true density is misspecified: Student's, skewed Student, Weibull, chi-squared, F distribution, etc.
- For consistency of mean regression parameters, mean regression function must be correctly specified.

Result.

Even if QML consistent, information matrix equality does **NOT** hold.

Therefore,

$$V_{\theta} = E \left[\frac{\partial s(z, \theta)}{\partial q'} \right]^{-1} E[s(z, \theta)s(z, \theta)'] E \left[\frac{\partial s(z, \theta)}{\partial q'} \right]^{-1}$$

should be estimated as whole sandwich, “robustly”:

$$\hat{V}_{\theta}^{(3)} = \left(\frac{1}{n} \sum_{i=1}^n \frac{\partial s(z_i, \hat{\theta})}{\partial q'} \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n s(z_i, \hat{\theta})s(z_i, \hat{\theta})' \right) \left(\frac{1}{n} \sum_{i=1}^n \frac{\partial s(z_i, \hat{\theta})}{\partial q'} \right)^{-1}.$$

Expectation-Maximization Algorithm

EM Algorithm

Why do we need the EM algorithm?

Many nonlinear models become easier to estimate after introducing **latent variables**.

Suppose we observe data y , but there exists an unobserved variable y^* such that

$$y = g(y^*).$$

If y^* were observed, we could maximize the **complete-data likelihood**

$$L(\theta; y^*) = \log f(y^* | \theta).$$

However, we only observe y , so the likelihood is

$$L(\theta; y) = \log f(y | \theta)$$

The difficulty is that integrating out the latent variable can make direct maximum likelihood estimation difficult.

Main idea of EM:

- E-step: calculate expectations with respect to latent variables given observed data;
- M-step: maximize the expected complete-data likelihood.

EM Algorithm: notation

Let

$$y = g(y^*)$$

be the relationship between latent data y^* and observed data y .
The density of latent data is

$$f(y^*|\theta).$$

The complete-data log likelihood is

$$L(\theta; y^*) = \log f(y^*|\theta).$$

For example, in the probit model:

$$f(y^*|\theta) = \phi(y^*|X\beta, I_n).$$

Define the EM objective function:

$$Q(\theta, \theta_t; y) = E[L(\theta; y^*)|y, \theta_t] = E_{y^*|y, \theta_t}[L(\theta; y^*)].$$

EM Algorithm: monotonicity

The key property of EM is:

$$Q(\theta, \theta_t; y) > Q(\theta_t, \theta_t; y)$$

implies

$$L(\theta; y) > L(\theta_t; y).$$

Therefore, if we update parameters by

$$\theta_{t+1} = \arg \max_{\theta} Q(\theta, \theta_t; y),$$

then

$$L(\theta_{t+1}; y) \geq L(\theta_t; y).$$

Thus, EM provides an alternative way to obtain the maximum likelihood estimator when direct maximization is difficult.

Why does EM work?

Define

$$H(\theta, \theta_t; y) = Q(\theta, \theta_t; y) - L(\theta; y).$$

Using the definition of Q :

$$H(\theta, \theta_t; y) = E_{y^*|y, \theta_t}[L(\theta; y^*|y)].$$

The difference can be written as

$$H(\theta_t, \theta_t; y) - H(\theta, \theta_t; y) = D_{KL}(f(y^*|y, \theta_t) \| f(y^*|y, \theta)).$$

Since

$$D_{KL}(q \| p) \geq 0,$$

we have

$$H(\theta_t, \theta_t; y) \geq H(\theta, \theta_t; y).$$

Therefore,

$$L(\theta; y) - L(\theta_t; y) \geq Q(\theta, \theta_t; y) - Q(\theta_t, \theta_t; y).$$

EM Algorithm: two steps

At iteration t :

E-step

Using current parameter θ_t :

$$f(y^*|y, \theta_t)$$

is calculated.

Then compute

$$Q(\theta, \theta_t; y) = E_{y^*|y, \theta_t}[L(\theta; y^*)].$$

M-step

Update parameters:

$$\theta_{t+1} = \arg \max_{\theta} Q(\theta, \theta_t; y).$$

Repeat:

$$\theta_0 \rightarrow \theta_1 \rightarrow \theta_2 \rightarrow \dots$$

until convergence.

Probit Example: Model Setup

Consider the latent variable representation of the probit model:

$$y^* = X\beta + \epsilon,$$

where

$$\epsilon|X \stackrel{iid}{\sim} N(0, I_n).$$

Observed outcome:

$$y_i = I(y_i^* > 0).$$

The latent variable y^* is not observed.

However, if y^* were observed, estimation would be equivalent to normal linear regression.

EM treats y^* as missing data.

Probit Example: E-step

The complete-data log likelihood is

$$L(\beta; y^*) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} (y^* - X\beta)' (y^* - X\beta).$$

The E-step computes

$$Q(\beta, \beta_t; y) = E[L(\beta; y^*) | y, \beta_t].$$

Expanding the quadratic:

$$Q(\beta, \beta_t; y) = -\frac{1}{2} E(y^{*'} y^* | \beta_t, y) + \beta' X' E(y^* | \beta_t, y) - \frac{1}{2} \beta' X' X \beta + c.$$

Define

$$\mu(\beta_t, y) = E(y^* | \beta_t, y).$$

Probit Example: Conditional expectation

Because

$$y^*|X, \beta_t \sim N(X\beta_t, I_n),$$

and

$$y_i = I(y_i^* > 0),$$

the conditional distribution of y^* is a **truncated normal distribution**.

If

$$y = 1,$$

then

$$E(y^*|\beta_t, y = 1) = X\beta_t + \frac{\phi(X\beta_t)}{\Phi(X\beta_t)}.$$

If

$$y = 0,$$

then

$$E(y^*|\beta_t, y = 0) = X\beta_t - \frac{\phi(X\beta_t)}{1 - \Phi(X\beta_t)}.$$

Probit Example: General E-step

Combining the two cases:

$$\mu(\beta_t, y) = X\beta_t + \frac{\phi(X\beta_t)}{\Phi(X\beta_t)[1 - \Phi(X\beta_t)]} [y - \Phi(X\beta_t)].$$

The E-step is completed once we obtain

$$\mu(\beta_t, y) = E(y^* | \beta_t, y).$$

Probit Example: M-step

Using the μ notation:

$$Q(\beta, \beta_t; y) = c(\beta_t) + \beta' X' \mu(\beta_t, y) - \frac{1}{2} \beta' X' X \beta.$$

The constant term does not depend on β .

The first-order condition is:

$$X' \mu(\beta_t, y) - X' X \beta = 0.$$

Therefore,

$$\beta_{t+1} = (X' X)^{-1} X' \mu(\beta_t, y)$$

The M-step is equivalent to an OLS regression:

$$\mu(\beta_t, y) \text{ on } X.$$

Probit Example: EM Algorithm

Putting everything together:

- 1 Choose starting value:

$$\beta_0.$$

- 2 Calculate:

$$\mu(\beta_0, y).$$

- 3 Regress

$$\mu(\beta_0, y)$$

on X to obtain

$$\beta_1.$$

- 4 Repeat:

$$\beta_1, \beta_2, \dots$$

until convergence.

Thus EM converts the probit estimation problem into repeated conditional expectation and least-squares updating.

Bootstrap and Bootstrap Inference

Bootstrap

Approaches to inference:

- ① Exact (finite sample): known distribution of data $\Rightarrow$ distribution of estimators/tests.
- ② Approximate: distribution of data unknown.
 - (a) Asymptotic approximation: use asymptotic theory to approximate distribution of estimators/tests by limits when sample size increases to infinity.
 - (b) Bootstrap approximation: approximate unknown distribution of data by its empirical distribution $\Rightarrow$ bootstrap distribution of estimators/tests.

Bootstrap idea: EDF

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n I\{x_i \leq x\}$$

approximates CDF of x , $F(x)$.

Bootstrap motivation:

$$|F_n(x) - F(x)| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Bootstrap principle: example

Consider example of $n = 2$:

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Parameter θ is regression coefficient of y on x ; $\hat{\theta}$ is OLS/analog estimator of θ :

$$\theta = \frac{E[xy]}{E[x^2]}$$

so

$$\hat{\theta} = \frac{x_1 y_1 + x_2 y_2}{x_1^2 + x_2^2} = \frac{1 \cdot 2 + 2 \cdot 1}{1^2 + 2^2} = \frac{4}{5}$$

(point estimate).

The empirical distribution of data is

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{cases} \begin{pmatrix} 1 \\ 2 \end{pmatrix} & \text{with probability } \frac{1}{2}, \\ \begin{pmatrix} 2 \\ 1 \end{pmatrix} & \text{with probability } \frac{1}{2}. \end{cases}$$

Suppose the data are drawn from this distribution. Then the two bootstrap observations

$$\begin{pmatrix} x_1^* \\ y_1^* \end{pmatrix}, \quad \begin{pmatrix} x_2^* \\ y_2^* \end{pmatrix}$$

can take four ordered combinations, each with probability $1/4$.

Bootstrap principle: example (continued)

Bootstrap distribution of data:

$$\left(\begin{pmatrix} x_1^* \\ y_1^* \end{pmatrix}, \begin{pmatrix} x_2^* \\ y_2^* \end{pmatrix} \right)$$

takes the following four values:

$$\begin{aligned} &\left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right), \quad \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right), \\ &\left(\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right), \quad \left(\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right), \end{aligned}$$

each with probability $1/4$.

Translate this to bootstrap distribution of $\hat{\theta}$:

$$\hat{\theta}^* = \begin{cases} \frac{1}{2} & \text{with probability } \frac{1}{4}, \\ \frac{4}{5} & \text{with probability } \frac{1}{2}, \\ 2 & \text{with probability } \frac{1}{4}. \end{cases}$$

We can use this bootstrap approximation to do bootstrap inference: construct bootstrap confidence intervals and perform bootstrap hypothesis testing.

We did this “easily” for $n = 2$, for which there were 3 points in support of $\hat{\theta}^*$.

For general n , there will be approximately

$$n^n$$

points in support of $\hat{\theta}^*$, which is HUGE!

Bootstrap algorithm (bootstrapping)

Target: construct bootstrap distribution of $\hat{\theta}$.

- ① Select number of *bootstrap replications* B (usually 1000 suffices). For $b = 1, 2, \dots, B$:
 - Construct bootstrap sample $(z_1^*, z_2^*, \dots, z_n^*)_b$ by drawing z_i^* randomly with replacement from the original sample $(z_1, \dots, z_n)$.
 - For each bootstrap sample, compute the bootstrap statistic $\hat{\theta}_b^*$ in the same way as $\hat{\theta}$ is computed from the original sample.
- ② Use $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$ as the bootstrap distribution of $\hat{\theta}$.
 - Bootstrap mean:

$$E^*[\hat{\theta}] \approx \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^*.$$

- Bootstrap quantiles: sort $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$ in increasing order and use

$$q_{\alpha_1}^* \approx \hat{\theta}_{[B\alpha_1]}^*, \quad q_{1-\alpha_2}^* \approx \hat{\theta}_{[B(1-\alpha_2)+1]}^*.$$

Bootstrap bias correction

Suppose $\hat{\theta}$ is consistent but biased analog estimator of θ :

$$E[\hat{\theta}] \neq \theta.$$

Then its bias is

$$E[\hat{\theta}] - \theta.$$

Let us use bootstrap to estimate the bias. Note recentering!

Bootstrap analog of

$$E[\hat{\theta}] - \theta = E^*[\hat{\theta}^*] - \hat{\theta} = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^* - \hat{\theta}.$$

Bootstrap bias corrected estimator of θ is

$$\tilde{\theta} = \hat{\theta} - \widehat{\text{Bias}} = \hat{\theta} - \left(\frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^* - \hat{\theta} \right) = 2\hat{\theta} - \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^*.$$

Bootstrap bias corrected estimator is still biased but almost unbiased.

Bootstrap confidence intervals

We want to construct confidence interval for parameter θ using its estimator $\hat{\theta}$.

Recall asymptotic $1 - \alpha$ level confidence interval:

$$CI^{asy}(\theta) = \left[\hat{\theta} \pm se(\hat{\theta}) q_{1-\alpha/2}^{N(0,1)} \right].$$

Efron's CI. Bootstrap $\hat{\theta}: \{\hat{\theta}_b^*\}_{b=1}^B$, and obtain bootstrap quantiles $q_{\alpha/2}^{E*}, q_{1-\alpha/2}^{E*}$.

$$CI^E(\theta) = \left[q_{\alpha/2}^{E*}, q_{1-\alpha/2}^{E*} \right].$$

Poor approximation except when distribution of $\hat{\theta}$ is symmetric.

Percentile CI. Bootstrap $\hat{\theta} - \theta$, by $\{\hat{\theta}_b^* - \hat{\theta}\}_{b=1}^B$. Let the bootstrap quantiles be $q_{\alpha/2}^{%*}, q_{1-\alpha/2}^{%*}$.

$$CI^{%}(\theta) = \left[\hat{\theta} - q_{1-\alpha/2}^{%*}, \hat{\theta} - q_{\alpha/2}^{%*} \right].$$

Good approximation for actual quantiles. Note: no standard errors are used!

Symmetric percentile CI. Bootstrap $|\hat{\theta} - \theta|$, by $\{|\hat{\theta}_b^* - \hat{\theta}|\}_{b=1}^B$. Let $q_{1-\alpha}^{|%|*}$ be the bootstrap quantile.

$$CI^{|%|}(\theta) = \left[\hat{\theta} \pm q_{1-\alpha}^{|%|*} \right].$$

Even better approximation for actual quantiles. Note: no standard errors are used!

Bootstrap confidence intervals (continued)

Percentile-t CI. Bootstrap $\frac{\hat{\theta} - \theta}{se(\hat{\theta})}$, by $\left\{ \frac{\hat{\theta}_b^* - \hat{\theta}}{se_b^*(\hat{\theta})} \right\}_{b=1}^B$. Let bootstrap quantiles be $q_{\alpha/2}^{*t}$, $q_{1-\alpha/2}^{*t}$.
 $CI^{*t}(\theta) = [\hat{\theta} - se(\hat{\theta})q_{1-\alpha/2}^{*t}, \hat{\theta} - se(\hat{\theta})q_{\alpha/2}^{*t}]$.

Very good approximation for actual quantiles. But: needs standard errors!

Symmetric percentile-t CI. Bootstrap $\left| \frac{\hat{\theta} - \theta}{se(\hat{\theta})} \right|$, by $\left\{ \left| \frac{\hat{\theta}_b^* - \hat{\theta}}{se_b^*(\hat{\theta})} \right| \right\}_{b=1}^B$.

Let bootstrap quantile be

$$q_{1-\alpha}^{*|t|}.$$

Then

$$CI^{*|t|}(\theta) = [\hat{\theta} \pm se(\hat{\theta})q_{1-\alpha}^{*|t|}].$$

Best approximation for actual quantiles. But: also needs standard errors!

Approximation quality

Quality of bootstrap approximations:

- Bootstrapping pivotal statistics (if one can pivotize) delivers better quality of approximation than bootstrapping non-pivotal statistics.
- Bootstrapping symmetrized statistics (if one can afford it) delivers better quality of approximation than bootstrapping non-symmetrized statistics.

Quality of bootstrap approximations compared to asymptotic approximations:

- **Asymptotic refinement:** Best bootstrap delivers better quality of approximation than asymptotics.
- But! You have to do bootstrap correctly (recentering, pivotization if possible, etc.).

Asymptotic refinement

Some asymptotic theory based on Edgeworth expansions.

For pivotal statistics,

$$\text{Error of asymptotic approximation} = \Phi(x) - F_{\hat{\theta}}(x) = O\left(\frac{1}{\sqrt{n}}\right),$$

while

$$\text{Error of bootstrap approximation} = F_{\hat{\theta}}^*(x) - F_{\hat{\theta}}(x) = O\left(\frac{1}{n}\right).$$

For non-pivotal statistics,

$$\text{Error of asymptotic approximation} = \Phi(x; V_{\theta}) - F_{\hat{\theta}}(x) = O\left(\frac{1}{\sqrt{n}}\right),$$

and

$$\text{Error of bootstrap approximation} = F_{\hat{\theta}}^*(x) - F_{\hat{\theta}}(x) = O\left(\frac{1}{\sqrt{n}}\right).$$

Resampling in time series models

Cannot do random resampling in time series! Why?

Instead of resampling observations, bootstrap blocks of observations.

1. Overlapping block bootstrap.

Let $\{(x_t, y_t)\}_{t=1}^T$ be the original sample and ℓ the block length.

First block:

$$\{(x_1, y_1), \dots, (x_\ell, y_\ell)\}.$$

Second block:

$$\{(x_2, y_2), \dots, (x_{\ell+1}, y_{\ell+1})\}.$$

Continue until the $(T - \ell + 1)$ th block. Then, blocks are drawn randomly with replacement from this set of blocks.

2. Non-overlapping block bootstrap.

First block:

$$\{(x_1, y_1), \dots, (x_\ell, y_\ell)\}.$$

Second block:

$$\{(x_{\ell+1}, y_{\ell+1}), \dots, (x_{2\ell}, y_{2\ell})\}.$$

Continue similarly. Then, blocks are drawn randomly with replacement from this set of blocks.

3. Stationary block bootstrap.

Variable block length. Set $p \in (0, 1)$. With probability $(1 - p)$ include next observation; with probability p end current block. Then, blocks are drawn randomly with replacement from this set of blocks.